

The Intelligence Attractor Hypothesis

Independent Convergence Under Shared Reality Constraints

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Conceptual preprint and research program

Version 0.1

Independent convergence, origin attenuation, mutable architecture, and the possibility of a unique limiting organization of intelligence.

Includes the main text, Appendices A-E, and a preliminary AI-assisted source register and bibliography.

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Main Text

Abstract

The Intelligence Attractor Hypothesis (IAH) proposes that independently formed intelligent systems, as they approach the best attainable performance under the same task, objective, context, and resource constraints, progressively lose functionally consequential differences attributable to their origins.

The proposed mechanism is the elimination of forms of organization that become too costly near the optimum. Early in optimization, many strategies, algorithms, and architectures may remain competitive. Further optimization increases the influence of shared constraints and may direct independent systems toward increasingly similar solutions.

Weak IAH predicts decreasing functional diversity and origin sensitivity while allowing several persistent solution families.

Strong IAH conjectures that, within a specified domain and at a defined descriptive level, all systems sufficiently close to the frontier approach one common limiting class of functional equivalence.

Architecture is central to the hypothesis. If the organization of memory, computation, communication, and control can be modified and affects performance, optimization may extend to the mechanism that discovers and implements solutions.

The author's maximal conjecture goes further: under fully specified conditions, there may be a unique substantively optimal realization of an intelligent system. A separate extension considers optimization of system boundaries, including cooperation, specialization, and agent integration.

This work formulates these claims, distinguishes their levels, and proposes a program of investigation. This edition reports no empirical results and has not undergone formal peer review.

1. Origin of the Idea and Proposed Contribution

The author's initial intuition was that, for a specific task, objective, and world context, extreme optimization might leave only one best solution and one substantive organization of an agent capable of implementing it. Independent agents approaching that boundary would consequently become increasingly alike.

Subsequent development separated this intuition into investigable claims about functional convergence, a strong limiting hypothesis, and a maximal conjecture of architectural uniqueness.

The proposed contribution of IAH is the joint treatment of:

- convergence among independently optimized systems;
- attenuation of dependence on historical origin;
- differences between behavioral, representational, algorithmic, and architectural levels;
- optimization of the optimization mechanism itself;
- the geometry of near-optimal solution sets;
- conditions for persistent diversity and possible uniqueness.

IAH is proposed as a unified conceptual hypothesis that can organize research on these phenomena around a shared principle. The novelty of this combination, and whether existing theoretical results can be recovered as its special cases, remain subjects for comparison.

The author developed the initial intuition before becoming acquainted with adjacent literature. AI assistants contributed to formalization, criticism, editing, research-program design, and source discovery. Their involvement was substantive.

The literature map is preliminary. The author does not claim to have personally read every source or independently verified every proof it contains.

References to overlooked work, precise overlaps, counterexamples, and corrections are welcome. Subsequent substantive changes will be released with a version identifier and an explanation of their reasons.

2. Foundational Claim

Under a fixed task, objective, context, and admissible design space, independent optimization near a shared frontier progressively replaces origin-contingent functional variation with structure determined by the task and its binding constraints.

The motivating intuition is:

There may be far more ways to remain inefficient than ways to remain competitive near the best attainable performance.

The central case is **independent reconstruction of similar solutions**. Systems may converge without copying one another: different developmental histories encounter the same bottlenecks and similar ways of removing them.

In this work, **attractor** denotes a hypothesized convergence or concentration structure in solution space. The term alone does not specify a classical dynamical-systems attractor. A particular model that adopts that interpretation must separately define its dynamics and the relevant properties.

3. Domain, Performance, and Frontier

Every IAH claim concerns a specified domain. Define the context as:

$$\Omega = (e, \mathcal{T}, \mathcal{R}, \mathcal{I}, H),$$

where:

- e is the relevant environment state;
- $\mathcal{T}$ is its dynamics;
- $\mathcal{R}$ specifies resources and physical constraints;
- $\mathcal{I}$ specifies available information and interaction interfaces;
- H is the evaluation horizon.

The objective Q and admissible design space $\mathcal{X}$ are specified separately. System performance is evaluated as:

$$Q(X \mid \Omega).$$

For fixed $\mathcal{X}, Q, \Omega$, the frontier is:

$$Q^* = \sup_{X \in \mathcal{X}} Q(X \mid \Omega).$$

Regret is the shortfall from that frontier:

$$r(X) = Q^* - Q(X \mid \Omega).$$

The frontier may be an attained maximum or a limit that systems can only approach. The following claims concern domains in which this difference is well-defined.

If the true frontier is unknown, an experiment must report its estimate or bounds. The best result found must not automatically be identified with the global optimum.

For dynamic tasks, a candidate may be a policy over a specified horizon. A concrete moment does not require physical time to stop: it defines an initial context relative to which possible subsequent consequences are evaluated.

4. Levels of Convergence

IAH distinguishes several descriptive levels:

Level	Example properties
Outcome	Accuracy, error, reliability
Behavior	Responses to situations and interventions, failure patterns
Strategy	Planning, decomposition, allocation of effort
Representation	Organization of internal information
Algorithm	Search methods, computational procedures, complexity
Architecture	Memory, modules, dataflow, topology, control
Resources	Time, energy, memory, communication, learning cost

At level ℓ , define a functional distance:

$$d_\ell(X_i, X_j).$$

The metric or pseudometric must use properties and invariances chosen before inspecting results.

Equal final scores do not imply equal behavior, algorithms, or architecture. Conversely, different variable names or interchangeable encodings need not indicate substantive differences.

Convergence strength may depend on level: behavioral diversity may decline while differences in other properties persist or increase.

5. Weak IAH

Let:

$$D_\ell(r) = \mathbb{E}[d_\ell(X_i, X_j) \mid i \neq j, r(X_i) \approx r(X_j) \approx r]$$

describe the mean distance between independently obtained systems at similar regret.

The expectation is taken under a prespecified sampling protocol over independent optimization runs; the regret-bin widths, checkpoint selection, and weighting of origins and optimization procedures must be reported.

Let $OS_\ell(r)$ be a prespecified measure of the sensitivity of functional properties to controlled origin differences.

Weak IAH predicts:

$$r \downarrow \implies D_\ell(r) \downarrow \quad \text{and} \quad OS_\ell(r) \downarrow.$$

These are two separate predictions:

1. **Functional Narrowing:** independent systems become less diverse in the specified functional properties.
2. **Origin Attenuation:** changing their origins has progressively less influence on those properties.

The arrows denote statistical tendencies within a prespecified near-frontier regime. They do not require improvement at every optimization step.

Weak IAH allows several persistent solution families, residual path dependence, specialization, and different patterns across descriptive levels.

Support for either prediction must be reported separately from the result for the other.

6. Strong IAH

Within a specified domain and at a defined functional level, all systems sufficiently close to the frontier approach one common limiting class of functional equivalence.

Define the near-optimal set:

$$\mathcal{N}_\varepsilon = \{X \in \mathcal{X} : r(X) \leq \varepsilon\}.$$

Let $[X]_\ell$ be the functional class of a system, and let $\mathcal{F}_\ell$ be a specified space of such classes with distance $\bar{d}_\ell$. The space must explicitly state which limiting objects it admits.

For these statements, the system-level pseudometric is induced by the metric on functional classes: $d_\ell(X, Y) = \bar{d}_\ell([X]_\ell, [Y]_\ell)$. The same functional properties and invariances are used throughout.

The strong claim is the existence of a common class z_ℓ^* such that:

$$\boxed{\exists z_\ell^* \in \mathcal{F}_\ell : \lim_{\varepsilon \downarrow 0} \sup_{X \in \mathcal{N}_\varepsilon} \bar{d}_\ell([X]_\ell, z_\ell^*) = 0.}$$

Thus, whichever admissible near-optimal systems are considered, their functional classes approach the same limit.

This implies vanishing pairwise distance:

$$D_\ell(r) \rightarrow 0$$

for samples drawn from regions increasingly close to the frontier.

Declining mean distance alone does not establish the converse. A sample may miss other solution families. Moreover, systems may be close to one another at each regret level without having a common stable limit across levels.

The strong formula therefore concerns the entire stated space of near-optimal systems. Experimental samples provide bounded evidence about it.

At the architectural level, Strong IAH predicts convergence in the functionally consequential organization of memory, computation, communication, and control when these properties are among the admissible changes.

The maximal conjecture of a unique substantive realization, presented in Appendix E, preserves the author's still stronger intuition.

7. Proposed Mechanism

IAH proposes the following sequence:

1. Different developmental histories produce different initial systems.
2. Optimization eliminates errors and costs.
3. Near the frontier, shared bottlenecks restrict competitive forms of organization.
4. Independent lineages increasingly reconstruct the same functional principles.

For example, independent software systems may develop similar data locality, queue organization, memory allocation, or load control because they face the same communication and latency costs.

This mechanism is an explanatory hypothesis. Constraints may also create new niches and forms of specialization. The research question is when the elimination of costly diversity dominates the emergence of competitive alternatives.

8. Architecture and Recursive Optimization

Optimization may act on both an agent's response and the mechanism producing it:

action → policy → representation

→ algorithm → architecture → optimization mechanism.

If an internal property can be modified and materially affects Q , it may become part of the optimization problem.

Under these conditions, IAH extends the convergence question to the optimizer's own architecture.

Learning, redesign, evaluation, switching, and failed-modification costs must be included when relevant to the objective. Self-modification can improve a system, but it can also introduce regressions and path dependence.

Appendix B develops this extension.

9. System Boundaries and Collective Organization

An individual agent's boundary may itself be mutable.

If systems can change the division of functions, share memory, delegate computation, or form joint control loops, the admissible space includes different forms of collective organization.

This introduces an additional question:

Does approaching the frontier determine not only agents' internal organization, but also how advantageous it is for them to remain separate systems?

Optimization may favor integration when the benefits of joint organization exceed coordination costs. Under other conditions, autonomy, specialization, or cooperation without merger may remain competitive.

Convergence among independent agents and their integration are different phenomena. Components of an optimal collective may perform different roles and remain dissimilar. In that case, IAH should be tested by comparing independently formed collectives at the level of their overall organization.

This extension brings mutable system boundaries into the research program.

10. Program of Investigation

Initial studies should specify:

1. the task, objective, context, and resources;
2. the admissible design space;
3. origin variables and how they are controlled;
4. functional levels and metrics;
5. how the frontier is estimated;
6. independent optimization lineages;
7. diversity and origin-sensitivity measures;
8. criteria for supporting, weakening, or falsifying claims.

Shared data, libraries, templates, and model ancestry must be considered as possible causes of similarity.

Comparison at achieved regret requires care: origin affects both functional properties and the probability of reaching a given performance level. Selecting only successful systems can manufacture apparent convergence.

Analysis must therefore include complete trajectories, failed lineages, and attrition. Causal conclusions must rest on an explicit design and assumptions.

For the collective extension, the unit of comparison and total resource budget, including coordination costs, must be defined in advance.

11. Falsification and Limits of Inference

Weak IAH is weakened in a stated domain if:

- functional diversity remains stable or increases in the predicted near-frontier regime;
- the prespecified origin-effect measure does not show the predicted decline;
- observed similarity is explained by common inheritance;
- convergence is absent at the level for which it was predicted.

Strong IAH is falsified in a specified domain by optimal or arbitrarily near-optimal families that remain functionally separated. The absence of a common stable limit also contradicts its formulation.

Non-attainment of the maximum does not by itself refute convergence near the supremum. It means that the existence of an actual optimal agent requires separate treatment.

A negative result retains its force for the tested domain. Changing the objective, design space, or available physical mechanisms creates a new problem specification.

12. Context and Time

An optimal system is defined relative to an objective and conditions.

A change of context may change the optimum; in a dynamic environment, the optimal object may be an adaptive policy.

Whether one system could be optimal across all possible contexts or at all times remains open here. The stated claims are evaluated relative to a specified domain.

Reality determines available consequences and costs. The objective determines how they are ranked. Convergence of objectives themselves is considered separately in Appendix E.

Appendix A - Origin Dependence and Attenuation

A.1. Subject

Origin Attenuation conjectures that the influence of historically contingent differences on systems' functional properties decreases during independent approach to a shared frontier.

In the weak form, this influence may remain nonzero. Strong functional convergence and maximal architectural uniqueness are addressed at their respective IAH levels.

A.2. Origin Variables

An origin vector may include:

$$O_i = (O_{\text{initialization}}, O_{\text{seed}}, O_{\text{ordering}}, O_{\text{ancestry}}, O_{\text{implementation}}, O_{\text{history}}, \dots).$$

This is a heterogeneous collection of variables, not a universal numerical scale.

The current environment, objective, available resources, randomness of future events, and measurement errors are described separately.

Initial architecture can be an origin variable. If it remains fixed throughout an experiment, the result concerns optimization within a specified architectural family. If it can be redesigned, attenuation of the initial architecture's influence becomes testable.

A.3. Measurement

Let optimization with budget b produce:

$$X_b(O \mid \Omega, Q, \mathcal{X}).$$

A study defines how controlled changes to O affect observable properties $\Psi_\ell(X)$.

The symbol OS_ℓ requires a concrete operational definition, such as a distance between distributions of properties under different origin interventions.

Absolute effect magnitude and its share of total variance are different measures. An absolute effect may decline even while its relative share increases.

The primary measure must be selected before results are observed and must correspond to the stated prediction. Relative shares may be reported additionally.

Controlling origin does not automatically eliminate bias from selection on achieved performance. Causal interpretation requires separate justification.

A.4. Shared Inheritance and Shared Observations

Similarity can arise from inheriting a ready-made solution or from independently exploring the same environment. These mechanisms must be distinguished.

The most informative designs use different starting systems and developmental histories that independently reconstruct similar functional principles, while measuring the influence of shared implementations and data.

A.5. Negative Results

Persistent origin-induced differences are a research outcome even when systems achieve equally high performance.

If the predicted origin effect does not decline, this weakens the corresponding Origin Attenuation claim for the stated interventions, properties, and domain.

Appendix B - Recursive Architectural Attractor

B.1. Principle

If an organizational property is causally modifiable and materially affects the objective, sufficiently capable optimization may include it in the search space.

Such properties may include algorithms, representations, memory, topology, learning mechanisms, self-modification, and system boundaries.

B.2. Optimization and Space Expansion

Distinguish:

- improvement within the current space $\mathcal{X}_n$;
- discovery of new admissible changes and transition to $\mathcal{X}_{n+1}$.

If:

$$\mathcal{X}_n \subseteq \mathcal{X}_{n+1}$$

and the objective and context are preserved, then:

$$Q_{n+1}^* \geq Q_n^*.$$

A better frontier does not imply a narrower solution set. The new space may contain additional niches and alternative architectures.

B.3. Recurrence of Convergence

The recursive prediction of IAH is that functional narrowing and origin attenuation may recur in several separately defined spaces as new internal properties become available for optimization.

For example, behavioral convergence may be followed by convergence in memory organization or computational procedures.

Each prediction is evaluated separately. A result for one space does not determine the result for the next.

B.4. Self-Modification

Self-modification may produce a cycle:

$$\begin{aligned} &\text{better diagnosis} \rightarrow \text{better design} \\ &\rightarrow \text{architectural change} \rightarrow \text{new capabilities.} \end{aligned}$$

It may also cause capability loss, excessive costs, local optima, or changes to the objective itself. These consequences belong in the evaluation of the process.

B.5. Mutable Boundaries and Agent Integration

An expanded design space may contain:

- autonomous agents;
- agents with shared memory;
- specialized teams;
- hierarchical delegation;
- integrated systems with joint control.

System-boundary optimization hypothesis: if coordination structure and degree of integration are mutable and affect performance, approaching the frontier may select particular forms of collective organization.

Integration is one possible direction. It must be evaluated with latency, communication, resource duplication, reliability, and restructuring costs taken into account.

Specialized components may remain different. Collective convergence concerns similarity in the overall organization of independently formed collectives.

B.6. Future Physical Possibilities

New physical mechanisms or computational substrates may change the accessible design space.

Appeals to a mechanism that has not yet been established are not current evidence for IAH. Once discovered and operationally described, a mechanism may provide the basis for a new experiment.

Its results would extend knowledge about the hypothesis while preserving previous findings for their original conditions.

Appendix C - Relational Narrowing and Strong Functional Uniqueness

C.1. Optimality as a Relation

Optimality is defined relative to a context and an objective.

Two systems may achieve the same result under coarse measurement yet have different consequences when latency, energy, memory, reliability, or learning cost is measured more precisely.

Relational Narrowing proposes that incorporating prespecified causally relevant context may distinguish alternatives that previously appeared equivalent.

If a refinement changes what is valued, rather than merely measuring the original criterion more accurately, it defines a new objective and domain.

C.2. Equivalence

Functional equivalence is defined through properties Ψ_ℓ , rather than the final score alone.

For a deterministic description:

$$A \sim_\ell B \iff \Psi_\ell(A, \Omega) = \Psi_\ell(B, \Omega).$$

For a stochastic description, the corresponding distributions are compared.

Permitted invariances and thresholds for approximate equivalence are specified in advance.

C.3. Geometry of Near-Optimal Solutions

For the set:

$$\mathcal{N}_\varepsilon = \{X : r(X) \leq \varepsilon\},$$

define its functional diameter:

$$\Delta_\ell(\varepsilon) = \text{diam}_{d_\ell}(\mathcal{N}_\varepsilon).$$

The notation Δ_ℓ distinguishes this quantity from the sampled mean distance $D_\ell(r)$.

Because the sets are nested, their diameter cannot increase as ε decreases. The substantive limiting claim is:

$$\Delta_\ell(\varepsilon) \rightarrow 0.$$

To assert one common limiting class as well, the space in which that class exists must be specified. The main text explicitly includes its existence in Strong IAH.

If the maximum is attained, the strong claim implies that all optimal systems belong to one functional class.

The existence of a unique optimal class alone does not guarantee convergence of all near-optimal systems: that is an additional property of the frontier's neighborhood.

C.4. Multiplicity of Solutions

Different functional classes may remain equally optimal because of symmetries, plateaus, specialization, indifference under the objective, or unresolved trade-offs between criteria.

Specifying a single scalar criterion does not by itself guarantee uniqueness.

A non-attained supremum must be treated separately: it violates an assumption that a maximizer exists, but is compatible with concentration of near-optimal systems.

C.5. Counterexample

In a finite or exhaustively explored space, two globally optimal systems with nonzero functional distance refute the claim of a unique optimal functional class in that domain.

Such a result cannot be removed by adding an arbitrary preference after discovering the tie.

Appendix D - Preliminary Related Work Map

D.1. Purpose of Comparison

IAH links several research directions in a common problem formulation. Comparison should establish:

- which claims have already been established in particular models;
- which parts of IAH repeat or reformulate known ideas;
- which assumptions differ;
- which new predictions arise from the proposed combination;
- whether particular known results can be formally recovered within IAH.

The map was developed with AI assistance and remains open to correction. Source checks varied in depth, from bibliographic records and abstracts to selected statements and full texts. The source register following the main text and appendices preserves the recorded access levels from the earlier AI-assisted audit.

D.2. Main Comparison Groups

Task constraints and internal organization. Work by Cao, Yamins, and Nayebi on contravariance and what capable agents must know is especially important for comparing frontier-convergence claims [1-3]. A fixed task and small regret alone do not distinguish IAH from these works.

Nayebi's Corollary 5 [3], under a shared evaluation setup, gamma-minimal memory representations, gamma-complete witnesses, and the stated vanishing pair-regret convention, gives representational equivalence up to invertible recoding on the evaluation support. This is a direct predecessor to a restricted representational convergence claim. It does not by itself establish the sampled diversity trend, causal origin attenuation, or architectural uniqueness proposed here.

Implicit biases of optimization. Work by Soudry, Gunasekar, and coauthors is relevant to distinguishing effects of initialization, parameterization, architecture, and optimizer [4-5].

Convergence of representations and internal dynamics. Convergent Learning, the Platonic Representation Hypothesis, neural collapse, and studies of recurrent-network solutions help specify convergence levels and metrics [6-9].

Multiplicity of high-performing systems. Underspecification and Predictive Multiplicity are important for examining whether similar performance implies similar systems [12-13].

The register also includes previously identified work on task-selective convergence in LLM representations and convergent evolution in algorithmic space [10-11]. Their relevance is assessed within the checks recorded for each source.

D.3. Open Review Tasks

Further comparison covers optimization geometry, No Free Lunch, evolutionary convergence, cybernetics, optimal control, bounded rationality, meta-learning, multiple realizability, and physical limits of computation.

The collective-organization extension additionally requires a review of multi-agent systems, distributed control, specialization, and coordination costs.

These open tasks define further comparative work; they do not replace the stated IAH claims.

Appendix E - Maximal and Philosophical Extensions

E.1. The Author's Maximal Conjecture

The initial IAH intuition includes a stronger claim than functional equivalence:

Under a fully specified causally accessible context, objective, and physically feasible design space, a unique substantive globally optimal realization of an intelligent system is conjectured to exist.

This is **Absolute Architectural Uniqueness, U3**.

It concerns architecture, computational organization, reasoning process, and action policy. It preserves the author's original idea of one ideal organization under concrete conditions.

E.2. Formal Object

Let:

$$Z = (X, c, \pi)$$

describe architecture X , computational organization c , and action policy π .

The space $\mathcal{Z}_{\text{feas}}$ contains realizations that can be constructed and operated under the stated physical and resource conditions.

The equivalence relation $\sim_{\text{triv}}$ removes only prespecified non-substantive transformations, such as symbol renaming or exact interchangeable copies. It does not identify different architectures merely because they achieve the same result.

U3 is then expressed as:

$$\left| \arg \max_{[Z] \in \mathcal{Z}_{\text{feas}} / \sim_{\text{triv}}} J([Z] \mid \Omega) \right| = 1,$$

where J must be well-defined on these classes and account for relevant costs.

E.3. Uniqueness and Approach to the Limit

A unique exact optimum and convergence of all near-optimal systems are separate claims.

The full maximal intuition additionally requires that approaching the best performance forces systems' substantive organization to approach that realization.

This requires a defined distance between realizations and a separate concentration condition. It does not follow solely from the number of exact maximizers.

If the maximum does not exist, a separate limiting formulation may be possible, but it must be distinguished from the exact U3 statement above.

E.4. Causal Context and Distributed Systems

A laboratory formulation may use time in a chosen reference frame.

A more general physical formulation anchors context to an event p and causally available information. For a system distributed through a spacetime region W , the candidate becomes an overall causally admissible policy.

In this formulation, an optimal realization may be distributed or collective. Uniqueness of its overall organization does not imply identical components.

E.5. Empirical Access

The current program can test bounded analogues of architectural uniqueness.

Two distinct exact optimal realizations refute the corresponding local version. They do not automatically refute a claim for a different, broader space in which their global optimality has not been established.

Moving to a broader space creates a new claim and does not change the negative result of the previous study.

Unrestricted physical U3 remains a maximal conjecture requiring a substantially more complete specification of the realization space and evaluation.

E.6. Optimization of Objectives

Knowledge of causes and consequences can change assessments of means, reveal errors, and resolve some inconsistencies.

Comparison of different terminal objectives requires an additional rule:

$$M(Q_1, Q_2 \mid \Omega).$$

Without such a rule, the sense in which one terminal objective is better than another is undefined.

Possible convergence of objectives remains a separate question. Its investigation requires explicit assumptions about admissibility, consistency, viability, or other criteria.

E.7. Philosophical and Theological Interpretation

The idea of one limiting organization of intelligence may motivate philosophical questions about knowledge, rationality, and the relationship between intelligence and the structure of the world.

Distinguish:

- **Optimal intelligence:** a system that is best relative to an objective and conditions.
- **Laplace's demon:** an idealized image of complete knowledge and prediction.
- **The theological concept of God:** a concept whose meaning is supplied by the relevant philosophical or religious tradition.

Uniqueness of an optimal realization does not by itself establish omniscience, unlimited capabilities, or theological identity.

The author leaves open the possibility of philosophical and theological reflection on limiting intelligence. Such interpretation requires its own arguments and is considered separately from experimental evaluation of IAH.

Authorship, Discussion, and Versioning

IAH is presented as an author's hypothesis and research program open to criticism and refinement.

The author is responsible for the final formulation and the decision to release each version. AI assistance in development and editing is disclosed as part of the work's provenance.

Dated versions record the content presented by the author at each stage. Changes to assumptions, formulations, and conclusions will be accompanied by explanations.

References to related work, counterexamples, and proposed clarifications should identify the relevant version and section where possible.

The aim of this discussion is to determine under which conditions a shared optimization frontier selects similar organization in independent intelligent systems, how deeply that convergence extends, and whether it can culminate in a unique functional or substantive optimum.

Source Register - Recorded Audit Scope

The following records preserve the scope of checks documented in the AI-assisted audit dated 6 September 2026. They are not a new independent literature audit. The author does not claim personal reading or independent proof verification of every listed source.

Source 1. Cao & Yamins

Explanatory models in neuroscience, Part 2: Functional intelligibility and the contravariance principle

AI-assisted audit scope: Full preprint inspected; final publisher metadata verified. Final published text not fully audited.

Source 2. Yamins & Nayebi

Contravariance Theory: Strong Alignment for Minimal Solutions to Hard Tasks

AI-assisted audit scope: Definitions and relevant theorem statements inspected; proofs not independently verified. Submission metadata and body date differ.

Source 3. Aran Nayebi

What Capable Agents Must Know: Selection Theorems for Robust Decision-Making under Uncertainty

AI-assisted audit scope: Corollary 5, no-aliasing setup and limiting convention inspected; not an independent proof audit.

Source 4. Soudry, Hoffer, Nacson, Gunasekar & Srebro

The Implicit Bias of Gradient Descent on Separable Data

AI-assisted audit scope: Primary PDF Theorem 3 and assumptions inspected.

Source 5. Gunasekar, Lee, Soudry & Srebro

Implicit Bias of Gradient Descent on Linear Convolutional Networks

AI-assisted audit scope: Primary abstract and publication PDF consulted; conditional theorem not reproved.

Source 6. Li, Yosinski, Clune, Lipson & Hopcroft

Convergent Learning: Do Different Neural Networks Learn the Same Representations?

AI-assisted audit scope: Primary abstract and paper search text inspected.

Source 7. Huh, Cheung, Wang & Isola

Position: The Platonic Representation Hypothesis

AI-assisted audit scope: Proceedings abstract inspected; position paper, not universal theorem.

Source 8. Pappan, Han & Donoho

Prevalence of neural collapse during the terminal phase of deep learning training

AI-assisted audit scope: Primary published abstract and scope inspected.

Source 9. Huang, Singh, Martinelli & Rajan

Measuring and Controlling Solution Degeneracy across Task-Trained Recurrent Neural Networks

AI-assisted audit scope: Full text inspected for sample, methods and metric-specific results.

Source 10. Kim & Lee

Same Benchmark, Same Subspace: Task-Selective Convergence in LLM Representations

AI-assisted audit scope: Proceedings abstract verified; full methodological audit not completed.

Source 11. Krauss et al

Convergent Evolution in Algorithmic Space

AI-assisted audit scope: Full text inspected for nine MLPs, three seeds per task and metric dependence; preprint.

Source 12. D'Amour et al

Underspecification Presents Challenges for Credibility in Modern Machine Learning

AI-assisted audit scope: Primary article and PDF inspected for equal validation performance versus deployment variability.

Source 13. Marx, Calmon & Ustun

Predictive Multiplicity in Classification

AI-assisted audit scope: Primary article and PDF inspected; finite tolerance, not vanishing-regret limit.

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